Test Name: KGiSL Hexaware Technical 3

Test Code: KGIS00113

Description:

Welcome to the test. This test consists of 1 section: Technical MCQ. Please read the instructions for each section carefully before starting.

Section Summary

Section	No. of Questions	Marks (+ / -)
Technical MCQ	40	+1 / -0

Technical MCQ Section

Duration: 60 minutes

Questions: 40

Marks per Question: +1 for correct, -0 for incorrect

1

Which constructor is correct?

A. BufferedReader br = new BufferedReader("file.txt");

B. BufferedReader br = new BufferedReader("file.txt");

C. BufferedReader br = new BufferedReader("file.txt");

D. BufferedReader br = new BufferedReader("file.txt");

Correct Answer:

B. BufferedReader br = new BufferedReader("file.txt");

Explanation:

BufferedReader wraps FileReader.

Marks: +1 for correct answer, -0 for wrong answer

2

How to make a thread daemon?

A. setBackground(true)

B. setDaemon(true)

C. makeDaemon()

D. daemon(true)

Correct Answer:

B. setDaemon(true)

Explanation:

setDaemon(true) must be called before start().

Marks: +1 for correct answer, -0 for wrong answer

3

What is the output of the following pseudocode?

```
Let arr = [1, 3, 6, 10, 15]
Set found = false
For i from 0 to length of arr - 1:
For j from i+1 to length of arr - 1:
If arr[j] - arr[i] == 5:
found = true
If found, print "YES", else print "NO"
```

A. YES

B. NO

C. ERROR

D. NULL

Correct Answer:

A. YES

Explanation:

The code checks if there is any pair of elements where the difference is 5.

• arr=6, arr=1 → 6-1=5

• arr=15, arr=10 → 15-10=5

So, at least one pair exists, and the output is "YES".

Marks: +1 for correct answer, -0 for wrong answer

4

What does this code print?

```
a ← 3
b ← 2

FUNCTION modify(x, y)
  x ← x + 1
  y ← y + 2
  PRINT x, y

modify(a, b)
PRINT a, b
```

A. 4 4 then 3 2

B. 3 2 then 4 4

C. 4 4 then 4 4

D. 3 2 then 3 2

Correct Answer:

A. 4 4 then 3 2

Explanation:

Local x and y change but original a, b stay unchanged (pass-by-value)

Marks: +1 for correct answer, -0 for wrong answer

5

Observe the Pseudo code and predict the output:

```
FUNCTION mix(n)
  IF n <= 0 THEN
    RETURN 0
  ELSEIF n MOD 2 = 0 THEN
    RETURN mix(n-1) + n/2
  ELSE
    RETURN mix(n-2) + n
PRINT mix(5)
```

A. 8

B. 9

C. 10

D. 11

Correct Answer:

B. 9

Explanation:

Explanation: mix(5)=mix(3)+5 → mix(3)=mix(1)+3 → mix(1)=mix(-1)+1=0+1=1 → total=1+3+5=9

Marks: +1 for correct answer, -0 for wrong answer

6

Observe pseudo code and predict the output:

```
FUNCTION alt(a, b)
  IF a <= 0 THEN
    RETURN b
  ELSE
    RETURN alt(a - 1, b + a)
PRINT alt(4, 1)
```

A. 10

B. 11

C. 12

D. 13

Correct Answer:

B. 11

Explanation:

Explanation: b incrementally grows: 1 + 4 + 3 + 2 + 1 = 11.

Marks: +1 for correct answer, -0 for wrong answer

7

Observe the code snippet and predict the output:

```
i ← 2
While i <= 8
  If (++i + i++) = i Then
    Print i
  End If
End While
```

A. 4 6 8

B. 5 7

C. Only 6

D. No output

Correct Answer:

D. No output

Explanation:

Explanation:

After evaluating the expression, it does not matches to 'i' in each iteration. So, it prints out nothing.

Marks: +1 for correct answer, -0 for wrong answer

8

Observe the code snippet and predict the output:

```
c ← '0'
Switch(c)
  case '0':
    Print "O"
    m ← '9'
    Switch(m)
      default: Print "X"
    End Switch
  default: Print "E"
  break
End Switch
```

A. O X E

B. O X

C. X E

D. O

Correct Answer:

A. O X E

Explanation:

Explanation:

'0' prints "O". Inner prints "X". Falls to outer default → "E".

Marks: +1 for correct answer, -0 for wrong answer

9

Observe the code snippet and predict the output:

```
Function funn(Integer a, Integer b)
 If((a mod b || b mod (a-1)) && (a ^ b < b)) Then
   a = (a ^ b) - 1
 Else
   Return (a - b) + 2
 End If
 Return a + b + 1
End Function

 Print funn(6, 3)
```

A. 5

B. 9

C. 7

D. 10

Correct Answer:

A. 5

Explanation:

a mod b = 0 (FALSE), b mod (a-1) = 3 mod 5 = 3 (TRUE), so left side is TRUE, but (a ^ b < b) → 6 ^ 3 = 5, 5 < 3 is FALSE → IF fails; return (6 - 3) + 2 = 5,

Marks: +1 for correct answer, -0 for wrong answer

10

Observe the pseudo code and predict it:

```
p ← 13
q ← 6
r ← 3

temp ← (p >> 2) ^ (q << 1)     // mix shifts + XOR
mask ← ( (p & q) << 1 ) | (r >> 1)

If( (temp & mask) > ( (p ^ r) << 1 ) ) Then
    p ← (temp | r) - (q >> 1)
Else
    q ← (mask ^ p) + (r << 2)
End If

Print p + q - mask
```

A. 14

B. 20

C. 22

D. 24

Correct Answer:

B. 20

Explanation:

Explanation:

p ← 13   // 1101

q ← 6   // 0110

r ← 3   // 0011

temp ← (p >> 2)⟶3 ^ (q << 1) ⟶ 12 // 15

mask ← ((p & q) << 1) ⟶8 | (r >> 1)⟶1 // 9

In if statement, (temp & mask) ⟶ 9 > ((p ^ r) ⟶14 << 1) ⟶28), evaluates to false.

Else block executes:

q ← (mask ^ p)⟶4 + (r << 2) ⟶ 12 // 16

prints out: p + q - mask = 13 + 16 - 9 = 20

Marks: +1 for correct answer, -0 for wrong answer

11

Observe the pseudo code and predict it:

```
a ← 14
b ← 6

Switch((a OR 3) - (b >> 1))
  Case 10: a ← a - b
  Case 11: a ← a + 2
  Case 12: a ← a * 2
  Case 13: a ← a - 1
  Default: a ← a + b
End Switch

Print a
```

A. 28

B. 15

C. 18

D. 33

Correct Answer:

D. 33

Explanation:

Explanation :

a ← 14

b ← 6

a OR 3 = 14 OR 3 = 15 (1110 | 0011 = 1111).

b >> 1 = 6 >> 1 = 3.

Difference = 15 - 3 = 12, so:

Case 12: a = 14 * 2 = 28.

Case 13: a ← a - 1 = 27

Default: a ← a + b = 27 + 6 = 33

Output = 33.

Marks: +1 for correct answer, -0 for wrong answer

12

Predict the output(array is 1-indexed):

```
A = [4, 9, 1, 7, 9]
SET max1 = 0
SET max2 = 0
FOR i = 1 TO 5
    IF A[i] > max1 THEN
        max2 = max1
        max1 = A[i]
    ELSE IF A[i] > max2 THEN
        max2 = A[i]
    ENDIF
ENDFOR
PRINT max2
```

A. 7

B. 9

C. 4

D. 1

Correct Answer:

B. 9

Explanation:

The trap is the duplicate 9. When the second 9 arrives: 9 > max1 (9) is false, but 9 > max2 (7) is true, so max2 becomes 9. The code doesn't exclude duplicates, so "second largest" ends up 9, not 7.

Marks: +1 for correct answer, -0 for wrong answer

13

Predict the output:

```
S = "Code"
SET T = ""
FOR i = 1 TO 4
  IF (i mod 2) = 1 THEN
    T = T + UPPER(S[i])
  ELSE
    T = T + LOWER(S[i])
  ENDIF
ENDFOR
PRINT T
```

A. CODE

B. cOdE

C. CoDe

D. code

Correct Answer:

C. CoDe

Explanation:

Odd positions are uppercased, even positions lowercased: 'C'→'C', 'o'→'o', 'd'→'D', 'e'→'e' → "CoDe". Answer (b) inverts the odd/even rule.

Marks: +1 for correct answer, -0 for wrong answer

14

Predict the output.

```
class Test:
    def init(self,x):
        self.x = x
t = Test(5)
print(t.x)
```

A. 5

B. x

C. Error

D. None

Correct Answer:

A. 5

Explanation:

Instance variable initialized via constructor.

Marks: +1 for correct answer, -0 for wrong answer

15

Predict the output assuming file contains "ABCDE".

```
f = open("test.txt","r")
f.seek(2)
print(f.read(1))
f.close()
```

A. A

B. B

C. C

D. D

Correct Answer:

C. C

Explanation:

Pointer moves to index 2 → character C.

Marks: +1 for correct answer, -0 for wrong answer

16

Predict the output.

```
import re
print(bool(re.search(r"\s", "Hello World")))
```

A. True

B. False

C. Error

D. None

Correct Answer:

A. True

Explanation:

\s matches whitespace character.

Marks: +1 for correct answer, -0 for wrong answer

17

Predict the output.

```
s = "banana"
print(s.count("a"))
```

A. 2

B. 3

C. 4

D. Error

Correct Answer:

B. 3

Explanation:

Character a appears three times.

Marks: +1 for correct answer, -0 for wrong answer

18

Predict the output.

```
try:
  print(x)
except NameError:
  print("Undefined")
```

A. Error

B. Undefined

C. None

D. x

Correct Answer:

B. Undefined

Explanation:

Accessing undefined variable raises NameError.

Marks: +1 for correct answer, -0 for wrong answer

19

Identify the output.

```
d = {"a":1}
d["b"] = 2
print(len(d))
```

A. 1

B. 2

C. Error

D. None

Correct Answer:

B. 2

Explanation:

New key-value pair increases dictionary size.

Marks: +1 for correct answer, -0 for wrong answer

20

Predict the output involving nested lists.

```
lst = [[1,2],[3,4]]
print(lst[1][0])
```

A. 1

B. 2

C. 3

D. 4

Correct Answer:

C. 3

Explanation:

Accesses second list then first element → 3.

Marks: +1 for correct answer, -0 for wrong answer

21

Predict the output of recursive execution.

```
def fun(n):
     if n == 0:
        return 0
     return n + fun(n-1)
print(fun(3))
```

A. 3

B. 5

C. 6

D. Error

Correct Answer:

C. 6

Explanation:

Calculation: 3 + 2 + 1 + 0 = 6.

Marks: +1 for correct answer, -0 for wrong answer

22

Consider nested loops below.

```
for i in range(2):
        for j in range(2):
                print(i, j)
```

How many times will print execute?

A. 2

B. 3

C. 4

D. 5

Correct Answer:

C. 4

Explanation:

Inner loop runs twice for each outer iteration → 2 × 2 = 4.

Marks: +1 for correct answer, -0 for wrong answer

23

Predict the output.

```
x = -1
if x:
print("True Block")
```

A. True Block

B. False Block

C. Error

D. No Output

Correct Answer:

A. True Block

Explanation:

Any non-zero number (including negative values) evaluates to True.

Marks: +1 for correct answer, -0 for wrong answer

24

Python evaluates expressions from left to right when operators have equal precedence.

What is the output?

```
print(20 / 5 * 2)
```

A. 2

B. 8.0

C. 4.0

D. 10

Correct Answer:

B. 8.0

Explanation:

Division and multiplication share precedence. Evaluation proceeds left to right → (20/5)=4, then 4*2=8.0.

Marks: +1 for correct answer, -0 for wrong answer

25

Python is often used as a scripting language for automation tasks such as file handling and system operations. Why is Python suitable for automation?

A. Requires extensive compilation

B. Limited library support

C. Simple syntax with OS integration modules

D. Hardware-level programming

Correct Answer:

C. Simple syntax with OS integration modules

Explanation:

Modules like os and sys allow easy automation of repetitive system tasks.

Marks: +1 for correct answer, -0 for wrong answer

26

Which method finds next matching subsequence?

A. matches()

B. find()

C. group()

D. split()

Correct Answer:

B. find()

Explanation:

find() searches for next occurrence.

Marks: +1 for correct answer, -0 for wrong answer

27

Which of the following is a valid method reference?

A. System.out.print

B. System.out::println

C. println::System.out

D. ::System.out

Correct Answer:

B. System.out::println

Explanation:

Syntax is ClassName::methodName.

Marks: +1 for correct answer, -0 for wrong answer

28

Which is fastest for random access?

A. LinkedList

B. ArrayList

C. HashSet

D. TreeSet

Correct Answer:

B. ArrayList

Explanation:

ArrayList supports O(1) random access using index

Marks: +1 for correct answer, -0 for wrong answer

29

Which of the following is NOT recommended in naming conventions?

A. Meaningful names

B. Short but unclear names

C. camelCase for methods

D. PascalCase for classes

Correct Answer:

B. Short but unclear names

Explanation:

Names should be meaningful and descriptive.

Marks: +1 for correct answer, -0 for wrong answer

30

Can abstract class have main method?

A. No

B. Yes

C. Only static

D. Only private

Correct Answer:

B. Yes

Explanation:

Abstract class can have main method and can be executed.

Marks: +1 for correct answer, -0 for wrong answer

31

Which modifier cannot be used with abstract class?

A. public

B. protected

C. final

D. static

Correct Answer:

C. final

Explanation:

final and abstract cannot be used together.

Marks: +1 for correct answer, -0 for wrong answer

32

Observe the code snippet and predict the output:

```
import java.util.*;
public class Test {
  public static void main(String[] args) {
    ArrayList<Integer> list = new ArrayList<>();
    list.add(1);
    list.add(2);
    list.add(3);
    for (Integer i : list) {
      list.remove(i);
    }
    System.out.println(list);
  }
}
```

A. []

B. [2]

C. [3]

D. ConcurrentModification Exception

Correct Answer:

D. ConcurrentModification Exception

Explanation:

Modifying an ArrayList while iterating using enhanced for-loop throws ConcurrentModification Exception.

Marks: +1 for correct answer, -0 for wrong answer

33

Observe the code snippet and predict the output:

```
class A {
    static void show() {
        System.out.print("A ");
    }
}

class B extends A {
    static void show() {
        System.out.print("B ");
    }
}

public class Main {
    public static void main(String[] args) {
        A obj = new B();
        obj.show();
    }
}
```

A. A

B. B

C. Compilation error

D. Runtime error

Correct Answer:

A. A

Explanation:

Static methods resolve using reference type, not object type.

Marks: +1 for correct answer, -0 for wrong answer

34

Java uses pass-by-value semantics for method calls, including object references. Consider the following program and determine how object references behave when passed to methods.

```
class Test {
  int x = 10;
  void modify(Test t) {
    t.x = 50;
    t = new Test();
    t.x = 100;
  }
  public static void main(String[] args) {
    Test obj = new Test();
    obj.modify(obj);
    System.out.println(obj.x);
  }
}
```

A. 10

B. 50

C. 100

D. Compilation error

Correct Answer:

B. 50

Explanation:

Reference copy is passed. Object field modified, but reassignment does not affect original

Marks: +1 for correct answer, -0 for wrong answer

35

Which statement correctly explains why Java avoids direct pointer manipulation unlike languages such as C or C++?

A. JVM does not support memory allocation

B. It improves security and prevents memory corruption

C. Java does not allow object creation

D. JVM manages only primitive variables

Correct Answer:

B. It improves security and prevents memory corruption

Marks: +1 for correct answer, -0 for wrong answer

36

Predict the output:

```
public class Main {
  public static void main(String[] args) {
    int i = 0;
    for (;;) {
      if (i == 3)
        break;
      System.out.print(++i + " ");
    }
  }
}
```

A. 1 2 3

B. 0 1 2

C. 1 2

D. infinite loop

Correct Answer:

A. 1 2 3

Explanation:

Pre-increment increases value before printing. Loop stops when i becomes 3.

Marks: +1 for correct answer, -0 for wrong answer

37

Predict output involving wrapper class caching behavior:

```
public class Main {
  public static void main(String[] args) {
    Integer a = 127;
    Integer b = 127;
    Integer c = 128;
    Integer d = 128;
    System.out.println(a == b);
    System.out.println(c == d);
  }
}
```

A. true

false

B. true

true

C. false

false

D. false

true

Correct Answer:

A. truefalse

Explanation:

Integer cache range is -128 to 127. Objects inside range reused. Outside range create new objects.

Marks: +1 for correct answer, -0 for wrong answer

38

Predict output involving multiple static blocks across inheritance:

```
class First {
  static {
    System.out.println("First static");
  }
}
class Second extends First {
  static {
    System.out.println("Second static");
  }
}

class Third extends Second {
  static {
    System.out.println("Third static");
  }
}

public class Main {
```

```
  public static void main(String[] args) {
    new Third();
  }
}
```

A. First static

Second static

Third static

B. Third static

Second static

First static

C. Second static

First static

Third static

D. First static

Third static

Second static

Correct Answer:

A. First staticSecond staticThird static

Explanation:

Static initialization occurs from topmost parent down to child class.

Marks: +1 for correct answer, -0 for wrong answer

39

Observe the Pseudo code and predict the output:

```
FUNCTION mix(x)
  IF x MOD 3 = 0 THEN
    RETURN x / 3
  ELSE
    RETURN mix(x + 1) + 1
  ENDIF

PRINT mix(5)
```

A. 2

B. 3

C. 4

D. 5

Correct Answer:

B. 3

Explanation:

Explanation: 5 → 6 → (6/3)=2 → +1 = 3.

Marks: +1 for correct answer, -0 for wrong answer

40

Observe pseudo code and predict the output:

```
sum ← 0
FOR i ← 1 TO 4
  FOR j ← 1 TO i
    IF (i + j) MOD 2 = 0 THEN
      sum ← sum + 1
PRINT sum
```

A. 4

B. 5

C. 6

D. 7

Correct Answer:

C. 6

Explanation:

Counts pairs where (i+j) even.

Pairs: (1,1),(2,2),(3,1),(3,3),(4,2),(4,4)=6.

Marks: +1 for correct answer, -0 for wrong answer